

Supplementary Information

Evaluating adaptive graph fusion for short-range weather forecasting with conditional local convolution

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S1 Scope, model inventory and evidence authority

This supplement documents 46 newly trained models: 40 primary models (four tasks, two variants, five seeds) and six additional mean-fusion ablations (two tasks, three seeds). Every model completed 100 epochs in FP32 with zero-inclusive training, validation selection and test scoring. All reported accuracy values originate from the new-training evidence package. Historical checkpoint rescoring, screening trials and previous zero-exclusive results do not supply values to these tables.

The Conditional Local Convolution Recurrent Network (CLCRN) and Conditional Local Convolution (CLConv) were introduced by Lin et al.[1]. The present contribution is an incremental adaptive graph-fusion (AGF) input extension and a controlled evaluation. The control is a matched local implementation: both variants use trainable node embeddings, whereas the upstream reference implementation registers those embeddings as a fixed buffer. The experiment therefore does not reproduce every upstream implementation choice.

The 40 primary replicates use seeds 2021–2025. The mean-fusion models use seeds 2023–2025 for temperature and humidity; the other 12 ablation entries reuse the corresponding control and AGF primary models. Thus the 18 ablation entries represent 18 selected models, of which only six are additional to the 40 primary models. The complete 46-row inventory, selected epochs, validation MAE, test scores, parameter counts, relative checkpoint paths and full SHA256 hashes are supplied in `data/model_inventory_46.csv`.

The accompanying main article reports the same 46-model evidence set. Main Figures 3–8 and Tables 1–4 share the definitions and checkpoints documented here. Main Figures 4 and 6 display MAE only; the primary tables and Supplementary Figure S2 retain both MAE and RMSE. Main Figures 5 and 7 show the five AGF models for each task; Supplementary Figures S6 and S5, respectively, preserve direct control–AGF comparisons. The conceptual main Figure 1 contains no measured experimental result, and main Figure 2 depicts the corrected pre-CLConv graph/residual fusion.

S2 Data, preprocessing and training protocol

The four WeatherBench-derived tasks are 2-m temperature, relative humidity, joint 10-m wind components (u, v), and total cloud cover. Hourly 2010–2018 fields occupy a regular 32×64 latitude–longitude grid (2,048 nodes). Each example has 12 input hours and 12 target hours. Window starts are separated by 24 hours, and the chronological, unshuffled split contains 2,300 training, 329 validation and 657 test examples. There is one fixed split; training seeds are not cross-validation folds.

Normalization moments are computed from the original training-input arrays, separately for each physical variable and each wind component. Both inputs and targets are cast to float32 before in-place subtraction and division by these moments. This ordering is frozen as `curated_float32_then_train_only_standardization_v1`. Validation and test data use training moments. Predictions and targets are inverse-transformed to physical units before scoring.

Table S1: Fixed protocol shared by every primary and ablation run.

Item	Specification
Training budget	100 epochs; early stopping disabled
Optimizer	Adam; learning rate 0.01; $\epsilon = 0.001$
Batch size	32 for training, validation and testing
Schedule	MultiStepLR at epochs 10, 20, 30, 40, 50; factor 0.95
Gradient clipping	Global gradient norm 5
Recurrent backbone	Two layers; 32 hidden units; one CLConv view
Geographic neighbours	25
Feature and backbone node embeddings	8 dimensions each
AGF node embedding / output width	10 / 8 dimensions
AGF retained neighbours	Top 4 per directed adjacency row
Decoder curriculum	Decay parameter 2,000
Additional context	No calendar features or periodic encoding
Precision	FP32; AMP and TF32 disabled
Checkpoint selection	Earliest minimum full-validation zero-inclusive MAE
Runtime recorded in training manifest	Python 3.13.7; PyTorch 2.10.0+cu128
GPU recorded in training manifest	NVIDIA GeForce RTX 5090 Laptop GPU; driver 610.88

Table S2: Training-input normalization moments recorded by the evaluator. Physical units apply to both mean and standard deviation.

Variable	Unit	Mean	Std. dev.	Raw dtype
Temperature	K	278.69592285	21.08552551	float32
Relative humidity	percentage points	78.13786316	18.10952377	float32
Surface wind (u)	m s^{-1}	-0.11169575	5.61085892	float32
Surface wind (v)	m s^{-1}	0.22041868	4.80938816	float32
Cloud cover	fraction	0.67634958	0.36162654	float32

Training and validation MAE include all finite targets, including valid zeros and negative values. A nonfinite prediction at a valid target raises an error. The new inference pass also rejects nonfinite error arrays instead of silently reducing the evaluation support. MAPE is not a primary metric. The raw target-zero audit is shown below; these zeros are observed values, not evidence of missing observations.

Table S3: Exactly zero raw targets by split. Wind combines its two output components.

Task	Split	Zero count	Total targets	Zero fraction (%)
Temperature	Train	0	56,524,800	0.00000
Temperature	Validation	0	8,085,504	0.00000
Temperature	Test	0	16,146,432	0.00000
Humidity	Train	0	56,524,800	0.00000
Humidity	Validation	0	8,085,504	0.00000
Humidity	Test	0	16,146,432	0.00000
Wind	Train	0	113,049,600	0.00000
Wind	Validation	0	16,171,008	0.00000
Wind	Test	0	32,292,864	0.00000
Cloud cover	Train	2,234,618	56,524,800	3.95334
Cloud cover	Validation	315,357	8,085,504	3.90028
Cloud cover	Test	626,840	16,146,432	3.88222

The cloud-cover training correction changes the optimization objective as well as model selection. Raw cloud zeros can acquire small floating-point residuals after standardization and inversion; zero-inclusive support is defined by finiteness and does not depend on exact inverse-transformed equality to zero. The absence of exact raw zeros in the other tasks does not remove the need for a consistent evaluator.

All trainable parameters are materialized before constructing the optimizer. The frozen training workflow audits optimizer membership and gradient coverage. Checkpoints include optimizer, scheduler and random-number-generator states for interrupted-run recovery. cuDNN deterministic mode is enabled and benchmarking is disabled; this does not establish bitwise equivalence across different hardware or software environments.

S3 Validation selection and individual model results

The selected checkpoint is the first epoch achieving the minimum MAE over the complete validation split among epochs 1–100. Test values do not enter selection. All 46 recorded histories contain epochs 1–100 exactly once; the inventory was checked against their earliest minima and against checkpoint/configuration/history SHA256 hashes. The following tables show the selected checkpoint and its cumulative 12-hour test scores. Validation MAE and test errors use the physical units indicated in each caption. Five seed labels do not imply paired stochastic streams across architectures.

Table S4: Temperature primary models (K). Every model trained for 100 epochs.

Variant	Seed	Selected epoch	Validation MAE	Test MAE	Test RMSE
Control	2021	90	1.2411	1.2543	2.0440
Control	2022	80	1.3251	1.3380	2.1234
Control	2023	92	1.1584	1.1691	2.0634
Control	2024	66	1.2534	1.2722	2.1440
Control	2025	92	1.2576	1.2563	2.1314
AGF	2021	67	1.3028	1.3252	2.2316
AGF	2022	100	1.3533	1.3797	2.1862
AGF	2023	87	1.2724	1.2862	2.0017
AGF	2024	78	1.1045	1.1091	1.9559
AGF	2025	90	1.3439	1.3548	2.2149

Table S5: Relative humidity primary models (percentage points). Every model trained for 100 epochs.

Variant	Seed	Selected epoch	Validation MAE	Test MAE	Test RMSE
Control	2021	100	4.5983	4.6677	7.2808
Control	2022	89	4.6034	4.6638	7.2816
Control	2023	82	4.5761	4.6394	7.1854
Control	2024	85	4.5756	4.6426	7.2712
Control	2025	66	4.5664	4.6263	7.2232
AGF	2021	97	4.4640	4.5222	6.9809
AGF	2022	85	4.4390	4.4986	6.9468
AGF	2023	78	4.4750	4.5348	7.0764
AGF	2024	87	4.4650	4.5279	6.9864
AGF	2025	94	4.5336	4.6024	7.0580

Table S6: Surface wind primary models (m s^{-1}). Every model trained for 100 epochs.

Variant	Seed	Selected epoch	Validation MAE	Test MAE	Test RMSE
Control	2021	78	1.2470	1.2441	1.9872
Control	2022	93	1.2382	1.2357	1.9607
Control	2023	98	1.2289	1.2258	1.9365
Control	2024	84	1.2547	1.2533	1.9908
Control	2025	89	1.2557	1.2512	1.9863
AGF	2021	86	1.2436	1.2392	1.9674
AGF	2022	82	1.2571	1.2548	2.0004
AGF	2023	98	1.2351	1.2334	1.9667
AGF	2024	84	1.2327	1.2309	1.9629
AGF	2025	94	1.3165	1.3201	2.1803

Table S7: Cloud cover primary models (fraction). Every model trained for 100 epochs.

Variant	Seed	Selected epoch	Validation MAE	Test MAE	Test RMSE
Control	2021	100	0.1472	0.1467	0.2494
Control	2022	99	0.1465	0.1458	0.2476
Control	2023	91	0.1501	0.1496	0.2523
Control	2024	91	0.1503	0.1496	0.2505
Control	2025	88	0.1485	0.1479	0.2501
AGF	2021	92	0.1475	0.1469	0.2499
AGF	2022	99	0.1494	0.1488	0.2508
AGF	2023	74	0.1472	0.1466	0.2440
AGF	2024	84	0.1485	0.1480	0.2507
AGF	2025	88	0.1521	0.1515	0.2495

Table S8: Six additional mean-fusion models; temperature is in K and humidity in percentage points.

Task	Seed	Selected epoch	Validation MAE	Test MAE	Test RMSE
Temperature	2023	89	1.2360	1.2544	1.9302
Temperature	2024	52	1.2059	1.2089	2.1240
Temperature	2025	98	1.3268	1.3382	2.0860
Humidity	2023	84	4.5342	4.5981	7.1015
Humidity	2024	93	4.5124	4.5763	7.0913
Humidity	2025	84	4.5603	4.6234	7.2037

S4 Primary statistics and exact-permutation sensitivity

Primary MAE and RMSE pool errors over all 657 test examples, 12 leads, 2,048 nodes and output components. With $e_a = \hat{y}_a - y_a$, M is the total number of scalar targets:

$$\text{MAE} = M^{-1} \sum_{a=1}^M |e_a|, \quad \text{RMSE} = \sqrt{M^{-1} \sum_{a=1}^M e_a^2}.$$

Aggregation uses double precision on differences formed from inverse-transformed FP32 tensors. RMSE is obtained from the pooled squared errors, not an average of per-window or per-lead RMSEs. The scalar tasks have 16,146,432 targets per model; wind has 32,292,864.

For each task/metric, the unit of statistical analysis is a trained model. I report group means and sample standard deviations ($n = 5$ per group), the difference $\Delta = \bar{x}_{\text{AGF}} - \bar{x}_{\text{control}}$, and the unadjusted two-sided 95% Welch interval. With $v_g = s_g^2/n_g$, the standard error is $\sqrt{v_A + v_C}$ and the Welch–Satterthwaite degrees of freedom are

$$v = \frac{(v_A + v_C)^2}{v_A^2/(n_A - 1) + v_C^2/(n_C - 1)}.$$

The interval is $\Delta \pm t_{0.975, v} \sqrt{v_A + v_C}$. Two-sided Welch p values receive Benjamini–Hochberg (BH) correction over all eight task/metric tests. Intervals are pointwise, not simultaneous confidence intervals.

Table S9: Primary unweighted estimates. Errors retain task-specific physical units; differences are AGF minus control.

Task	Metric	Control (mean $\pm$ s)	AGF (mean $\pm$ s)	Δ [95% CI]
Temperature	MAE	1.2580 $\pm$ 0.0603	1.2910 $\pm$ 0.1075	+0.0330 [−0.1004, +0.1664]
Temperature	RMSE	2.1012 $\pm$ 0.0445	2.1181 $\pm$ 0.1292	+0.0168 [−0.1408, +0.1745]
Humidity	MAE	4.6480 $\pm$ 0.0174	4.5372 $\pm$ 0.0389	−0.1108 [−0.1584, −0.0632]
Humidity	RMSE	7.2484 $\pm$ 0.0427	7.0097 $\pm$ 0.0550	−0.2387 [−0.3113, −0.1661]
Wind	MAE	1.2420 $\pm$ 0.0114	1.2557 $\pm$ 0.0372	+0.0137 [−0.0318, +0.0591]
Wind	RMSE	1.9723 $\pm$ 0.0234	2.0155 $\pm$ 0.0933	+0.0432 [−0.0711, +0.1576]
Cloud cover	MAE	0.1479 $\pm$ 0.0017	0.1484 $\pm$ 0.0020	+0.0005 [−0.0022, +0.0031]
Cloud cover	RMSE	0.2500 $\pm$ 0.0017	0.2490 $\pm$ 0.0028	−0.0010 [−0.0045, +0.0025]

The exact-permutation sensitivity uses the absolute difference in group means and enumerates all $\binom{10}{5} = 252$ allocations, including the observed allocation. Its two-sided p value is the fraction at least as extreme as observed (with 10^{-12} numerical

comparison tolerance). Exactness holds under exchangeable group labels, not the weaker equal-means null with arbitrary unequal variances. Its eight p values form a separate BH family; this sensitivity does not replace primary Welch inference. No paired sign test is used.

Table S10: Primary significance and permutation sensitivity; each pair of q columns uses its own eight-comparison family.

Task	Metric	Change (%)	Welch p	Welch q	Exact p	Exact q
Temperature	MAE	+2.625	0.570054	0.760072	0.555556	0.770975
Temperature	RMSE	+0.800	0.794234	0.794234	0.777778	0.777778
Humidity	MAE	-2.384	0.001510	0.006040	0.007937	0.031746
Humidity	RMSE	-3.294	0.000082	0.000653	0.007937	0.031746
Wind	MAE	+1.100	0.469944	0.760072	0.634921	0.770975
Wind	RMSE	+2.193	0.365690	0.760072	0.452381	0.770975
Cloud cover	MAE	+0.306	0.707722	0.794234	0.674603	0.770975
Cloud cover	RMSE	-0.408	0.510257	0.760072	0.619048	0.770975

Only humidity MAE and RMSE satisfy $q < 0.05$ in both families. Their mean reductions are 2.38% and 3.29%, respectively. For both humidity metrics the permutation p is $2/252 = 0.007937$ and the separately corrected q is 0.031746. The other six comparisons do not establish differences at the stated threshold. Failure to reject a null is not proof of equal performance. These tests quantify training-seed variation on the fixed split and do not estimate variation across climates, periods, splits or datasets.

S5 Latitude-weighted sensitivity

For latitude ϕ_i , I replace unit weights by $w_i = \cos \phi_i$ and calculate

$$\text{MAE}_w = \frac{\sum_a w_a |e_a|}{\sum_a w_a}, \quad \text{RMSE}_w = \sqrt{\frac{\sum_a w_a e_a^2}{\sum_a w_a}}.$$

The evaluator normalizes node weights to mean one; repeating them over windows, leads and channels is algebraically equivalent to the displayed denominator. This changes evaluation only, not training, validation selection or checkpoints. Its eight Welch tests constitute a separate sensitivity family. Humidity remains the only task with both corrected metrics below 0.05.

Table S11: Latitude-weighted scores, pointwise Welch intervals and separately corrected q values.

Task	Metric	Control mean	AGF mean	Δ [95% CI]	BH q
Temperature	MAE	1.1722	1.2383	+0.0661 [−0.0971, +0.2293]	0.659747
Temperature	RMSE	2.0135	2.0503	+0.0369 [−0.1136, +0.1873]	0.659747
Humidity	MAE	4.6573	4.5300	−0.1273 [−0.1910, −0.0637]	0.010764
Humidity	RMSE	7.2278	6.9485	−0.2793 [−0.3689, −0.1896]	0.000871
Wind	MAE	1.2191	1.2312	+0.0121 [−0.0331, +0.0572]	0.659747
Wind	RMSE	1.9173	1.9585	+0.0412 [−0.0729, +0.1553]	0.659747
Cloud cover	MAE	0.1595	0.1600	+0.0005 [−0.0024, +0.0034]	0.692620
Cloud cover	RMSE	0.2528	0.2518	−0.0009 [−0.0044, +0.0025]	0.659747

S6 Persistence on the same split

Persistence repeats the final observed input over all 12 forecast leads. It is evaluated once, with the same inverse transformation, test split, zero-inclusive target support and unweighted aggregation as the trained models. It has no training-seed standard deviation. Control and AGF entries below are means of their five trained replicates. The table is descriptive; persistence is not an additional primary hypothesis test. Historical published CLCRN scores have different evaluation provenance and are not included in this matched table.

Table S12: Current-split cumulative 12-hour error. Humidity units are percentage points; cloud cover is a fraction.

Task	Method	Training replicates	MAE	RMSE
Temperature	Persistence	–	1.4214	2.7001
Temperature	Control	5	1.2580	2.1012
Temperature	CLCRN-AGF	5	1.2910	2.1181
Humidity	Persistence	–	6.4248	10.4471
Humidity	Control	5	4.6480	7.2484
Humidity	CLCRN-AGF	5	4.5372	7.0097
Wind	Persistence	–	1.5372	2.5354
Wind	Control	5	1.2420	1.9723
Wind	CLCRN-AGF	5	1.2557	2.0155
Cloud cover	Persistence	–	0.1620	0.2725
Cloud cover	Control	5	0.1479	0.2500
Cloud cover	CLCRN-AGF	5	0.1484	0.2490

S7 Three-seed component ablation

The no-AGF condition removes the complete input augmentation and is exactly the matched control. Mean fusion retains the two AGF projections but uses $\mathbf{z} = (\mathbf{q} + \mathbf{r})/2$ instead of the learned gate. Full AGF uses $\mathbf{z} = \mathbf{g} \odot \mathbf{q} + (1 - \mathbf{g}) \odot \mathbf{r}$. Both branches use the same pre-CLConv input: $\mathbf{q}$ is graph-aggregated and $\mathbf{r}$ is a node-wise residual linear projection. Neither branch is a CLConv output. All conditions receive the same 100-epoch FP32 budget, and every row uses seeds 2023–2025.

Table S13: Aligned three-seed ablation. Values are mean $\pm$ sample standard deviation, not five-seed summaries.

Task	Variant	MAE	RMSE
Temperature	No AGF (control)	1.2325 ± 0.0555	2.1129 ± 0.0433
Temperature	Mean fusion	1.2672 ± 0.0656	2.0468 ± 0.1027
Temperature	Full AGF	1.2500 ± 0.1268	2.0575 ± 0.1382
Humidity	No AGF (control)	4.6361 ± 0.0086	7.2266 ± 0.0430
Humidity	Mean fusion	4.5993 ± 0.0236	7.1322 ± 0.0622
Humidity	Full AGF	4.5550 ± 0.0412	7.0403 ± 0.0475

Humidity has progressively lower mean errors from control to mean fusion to learned gating in this subset. Temperature is metric-dependent: control has the lowest mean MAE, while mean fusion has the lowest mean RMSE. The three-seed ablation is descriptive and does not establish that gating is universally necessary. Individual control/AGF values are reported above; the six additional models appear in Table S8. The complete 18-row aligned file is `data/ablation_per_seed.csv`.

S8 Missing-input diagnostic: 64 windows and three masks

The diagnostic uses 64 deterministic, approximately equally spaced test indices spanning the 657-window chronology, including indices 0 and 656. The exact indices are recorded separately for each task. All primary models share this subset and mask seeds 11, 22 and 33. For each test example and nonzero nominal ratio $r \in \{0.1, 0.2, 0.3, 0.4\}$, the evaluator samples $\lfloor 2048r \rfloor$ nodes without replacement. The example’s mask is held fixed over all 12 input hours and all input channels. Different examples receive different draws. For a given ratio/mask seed, the same masks are reused across architectures and training seeds.

Masked standardized inputs are set to zero, corresponding to the training mean in physical units. This is input replacement, not target masking, and is distinct from the valid zero-target protocol. Training is unchanged. At each ratio, three relative MAE increases are averaged within each trained seed; the displayed mean and sample standard deviation are then computed across the five trained seeds. Masks are nested technical repeats, not 15 independent training replicates. Each relative increase uses the same checkpoint’s clean MAE on the same 64-window subset:

$$100 \frac{\text{MAE}_{\text{masked},s,m} - \text{MAE}_{\text{clean},s}}{\text{MAE}_{\text{clean},s}}.$$

Table S14: Relative MAE increase (%) on the shared 64-window subset. At zero missingness every relative increase is zero by definition.

Task	Nominal ratio	Masked nodes	Control (mean $\pm s$)	AGF (mean $\pm s$)
Temperature	0.1	204	115.80 $\pm$ 5.96	114.32 $\pm$ 12.85
Temperature	0.2	409	233.37 $\pm$ 11.93	229.99 $\pm$ 25.54
Temperature	0.3	614	352.13 $\pm$ 17.85	346.55 $\pm$ 38.11
Temperature	0.4	819	474.09 $\pm$ 23.92	465.95 $\pm$ 50.67
Humidity	0.1	204	17.79 $\pm$ 0.31	18.01 $\pm$ 0.30
Humidity	0.2	409	35.87 $\pm$ 0.58	36.42 $\pm$ 0.61
Humidity	0.3	614	54.12 $\pm$ 0.82	55.18 $\pm$ 0.89
Humidity	0.4	819	72.99 $\pm$ 1.02	74.71 $\pm$ 1.19
Wind	0.1	204	20.29 $\pm$ 0.64	20.12 $\pm$ 0.22
Wind	0.2	409	41.47 $\pm$ 1.12	41.03 $\pm$ 0.70
Wind	0.3	614	63.44 $\pm$ 1.57	62.73 $\pm$ 1.39
Wind	0.4	819	85.75 $\pm$ 1.92	84.67 $\pm$ 2.28
Cloud cover	0.1	204	7.50 $\pm$ 0.41	7.89 $\pm$ 0.85
Cloud cover	0.2	409	15.86 $\pm$ 0.89	16.67 $\pm$ 1.50
Cloud cover	0.3	614	25.05 $\pm$ 1.48	26.45 $\pm$ 1.97
Cloud cover	0.4	819	35.49 $\pm$ 2.24	37.52 $\pm$ 2.29

AGF does not show uniform missing-input robustness: relative increases are slightly lower for temperature and wind, but higher for humidity and cloud cover. Relative degradation is not a comparison of absolute masked errors between models with different clean errors. Absolute clean/masked MAE and RMSE for all 520 records are supplied in `data/robustness_per_seed_mask.csv`. This sampled input-replacement diagnostic is not a full-test benchmark or a test of actual missing-observation patterns, and no robustness significance claim is made.

Main Figure 7 overlays the four AGF task curves using a common relative-change axis. Supplementary Figure S5 retains the control and AGF curves in separate task panels, using exactly the same checkpoints, windows, masks and nested aggregation as Table S14. It provides the comparison needed to interpret the AGF-only main figure; no additional models or masks are introduced.

S9 Full-test gate statistics

The gate is $\mathbf{g} = \sigma(W_g[\mathbf{r}||\mathbf{q}] + b_g)$ and its value weights the graph-aggregated branch in the fusion above. Statistics are collected from all scalar gate outputs before any channel averaging. Each AGF model contributes $657 \times 12 \times 2048 \times 8 = 129,171,456$ values; the five-model task pool has 645,857,280 values. All 100 histogram bins over $[0, 1]$ sum to the corresponding scalar count. A value displayed as 1.0000 can result from floating-point sigmoid saturation or rounding; it does not indicate a hard selection mechanism.

Table S15: Gate statistics for every AGF training seed. Standard deviation describes scalar gate dispersion within a model.

Task	Seed	Mean	Population SD	Minimum	Maximum
Temperature	2021	0.6156	0.2331	0.0014	0.9967
Temperature	2022	0.6230	0.2706	0.0010	1.0000
Temperature	2023	0.6260	0.1393	0.1331	0.9825
Temperature	2024	0.5789	0.1512	0.0427	0.9746
Temperature	2025	0.5573	0.1381	0.0640	0.9644
Humidity	2021	0.5853	0.2412	0.0018	0.9995
Humidity	2022	0.5685	0.2130	0.0088	0.9992
Humidity	2023	0.6644	0.2369	0.0164	0.9990
Humidity	2024	0.5236	0.2417	0.0018	0.9981
Humidity	2025	0.5107	0.2569	0.0014	0.9993
Wind	2021	0.4950	0.1133	0.0994	0.8989
Wind	2022	0.5045	0.1439	0.0705	0.9541
Wind	2023	0.5129	0.1210	0.0600	0.9177
Wind	2024	0.4954	0.1241	0.0845	0.9529
Wind	2025	0.5304	0.1427	0.0486	0.9418
Cloud cover	2021	0.4632	0.2173	0.0175	0.9850
Cloud cover	2022	0.4416	0.2180	0.0088	0.9970
Cloud cover	2023	0.5452	0.1870	0.0418	0.9799
Cloud cover	2024	0.4603	0.1958	0.0023	0.9763
Cloud cover	2025	0.4666	0.2044	0.0101	0.9702

Table S16: Gate moments pooled across five AGF seeds, with equal scalar support per model.

Task	Mean	Pooled population SD	Minimum	Maximum
Temperature	0.6001	0.1963	0.0010	1.0000
Humidity	0.5705	0.2445	0.0014	0.9995
Wind	0.5076	0.1302	0.0486	0.9541
Cloud cover	0.4754	0.2080	0.0023	0.9970

The pooled variance includes both within-model dispersion and differences among seed means; it is not the standard deviation of five seed means. The gate values are correlated over space and time and are not independent inferential samples. Their magnitudes do not prove physical interpretability, causal importance or forecast improvement. Both projections have learned scales, so a numerical gate coefficient alone does not measure a branch’s contribution to output error.

Main Figure 8 uses mirrored profiles interpolated through the recorded 100-bin histogram centres, with widths normalized separately for each task. Its boxes, median lines and whiskers denote approximate 25th–75th, 50th and 5th–95th percentiles, obtained by linear interpolation of cumulative bin counts. These are binned approximations because the package records histograms rather than individual scalar activations. The mean diamonds and reported population standard deviations use the exact streamed moments in Table S16, not moments inferred from bin centres. Supplementary Figure S4 retains the individual-seed and pooled densities from the same histogram records.

S10 Forecast-lead support and reproducibility boundaries

Exact-lead scores use only lead h ; cumulative scores pool leads 1 through h . They must not be interchanged. The machine-readable horizon file contains 480 rows (40×12), each with both definitions. At lead 12, every cumulative value agrees with its primary aggregate. The table below illustrates exact-lead scores at 6 and 12 hours; all leads remain available in `data/horizon_per_seed.csv`. No additional per-lead significance family is claimed.

Main Figure 5 plots AGF exact-lead MAE, cumulative MAE and exact-lead RMSE, each summarized over five training seeds. Supplementary Figure S6 shows exact-lead and cumulative MAE for both control and AGF. The latter supplies the matched-model context without confusing the main figure’s grey exact-RMSE curve with a control-model curve.

Table S17: Exact-lead scores (mean $\pm$ sample SD over five models), in the task-specific physical units.

Task	Variant	Lead (h)	Exact MAE	Exact RMSE
Temperature	Control	6	1.3388 ± 0.0776	2.1614 ± 0.0615
Temperature	Control	12	1.8639 ± 0.1216	2.8042 ± 0.1336
Temperature	AGF	6	1.3488 ± 0.0931	2.1260 ± 0.0900
Temperature	AGF	12	2.0530 ± 0.2591	2.9300 ± 0.2845
Humidity	Control	6	4.9726 ± 0.0206	7.3972 ± 0.0539
Humidity	Control	12	6.5617 ± 0.0295	9.2681 ± 0.0595
Humidity	AGF	6	4.7942 ± 0.0387	6.9943 ± 0.0730
Humidity	AGF	12	6.5154 ± 0.0749	9.2202 ± 0.0819
Wind	Control	6	1.2720 ± 0.0095	1.8948 ± 0.0193
Wind	Control	12	1.9213 ± 0.0247	2.7604 ± 0.0395
Wind	AGF	6	1.2857 ± 0.0337	1.9330 ± 0.0749
Wind	AGF	12	1.9446 ± 0.0745	2.8249 ± 0.1559
Cloud cover	Control	6	0.1555 ± 0.0019	0.2560 ± 0.0019
Cloud cover	Control	12	0.1927 ± 0.0026	0.3051 ± 0.0020
Cloud cover	AGF	6	0.1560 ± 0.0021	0.2550 ± 0.0028
Cloud cover	AGF	12	0.1933 ± 0.0032	0.3034 ± 0.0046

All 46 inference records were checked against their checkpoint, configuration, summary, training-history and run-manifest digests. The independent supplement audit recomputes means, sample standard deviations, Welch statistics and intervals, all 252 permutation allocations, BH corrections, nested robustness summaries and pooled gate moments. This audit validates the released records and calculations; it is not a new training experiment or an independent implementation of the network.

The maximum absolute difference between the new inference MAE/RMSE and the new training-summary metrics is 4.061×10^{-10} . Small differences arise from aggregation order; historical metrics are not used as the reference for this check.

The source snapshot and processed arrays specified by the data-source records are the operational reproduction boundary. Data-source JSON files record the SHA256 hashes of `trn.pkl`, `val.pkl`, `test.pkl` and `position_info.pkl`, together with dimensions and scaler moments. This manuscript package includes the recorded results; the large processed input arrays are distributed separately in the reproduction materials. This package does not establish a checksum-verified reconstruction from the original ERA5 archive, broader weather-model competitiveness, uncertainty under alternative data splits, or attribution to a particular physical process. Full retraining requires the processed input arrays, frozen training source, configurations and compatible accelerator environment. Running `validate_supplement_sync.py` in this supplement directory audits the local records and the figure/data synchronization. Running `pdflatex supplement.tex` twice rebuilds the PDF; neither action reruns inference or training.

No fresh training-throughput, inference-latency or memory benchmark is introduced here. Analysis wall-clock fields include evaluation and diagnostics and are not matched timing measurements. Any historical efficiency measurements discussed in the main article retain their separate provenance and must not be presented as measurements of these 46 new checkpoints.

S11 Evidence files accompanying this supplement

The accompanying data/ directory contains the following inspectable records. File digests are listed in `supplement_sources.json`; the synchronization audit is in `supplement_validation.json`. The source provenance manifest preserves the original experimental file identities.

- `model_inventory_46.csv`: the complete new-model inventory with selected validation epochs and full checkpoint hashes.
- `primary_per_seed.csv`, `table1_primary_statistics.csv`: 40 primary model values and eight unweighted comparisons, including Welch and exact-permutation statistics.
- `latitude_weighted_sensitivity.csv`: separately corrected spatial-weighting sensitivity.
- `ablation_per_seed.csv`, `table3_ablation_statistics.csv`: 18 aligned entries and six three-seed summaries.
- `table2_current_context.csv`, `persistence_*.json`: same-split persistence and model context.
- `raw_target_zero_fractions.csv`, `dataset_*.json`: raw-zero support, split dimensions, moments and input-file hashes.

- `horizon_per_seed.csv`: 480 exact-lead/cumulative records.
- `robustness_indices_*.json`, `robustness_per_seed_mask.csv`, `robustness_summary.csv`: exact 64-window subsets, 520 clean/masked records, and summaries respecting mask nesting.
- `gate_per_seed.csv`, `gate_histogram.csv`, `table4_gate_statistics.csv`: 20 model-level records, 100-bin histograms and pooled full-test gate statistics.
- `inference_*.json`: 46 new-checkpoint inference records, selected epochs, configuration/checkpoint/history hashes, metric sufficient sums and diagnostics where applicable.
- `verification_report.json`, `provenance_manifest.json`, `analysis_code_manifest.json`: original completion/selection checks, source identities and frozen analysis-code provenance.

Full-resolution supplementary figures follow. Figure S1 is conceptual; Figures S2–S6 are generated from the same new evidence described above. They do not introduce additional trained models or statistical families. Figure S2 preserves relative primary effects and Welch intervals; S3 shows the three current-split methods in physical units; S4 retains seed-level gate densities; S5 and S6 retain both-variant missing-input and forecast-lead comparisons.

References

- [1] Lin, H. et al. Conditional local convolution for spatio-temporal meteorological forecasting. *Proc. AAAI Conf. Artif. Intell.* **36**, 7470–7478 (2022).

S12 Supplementary figures

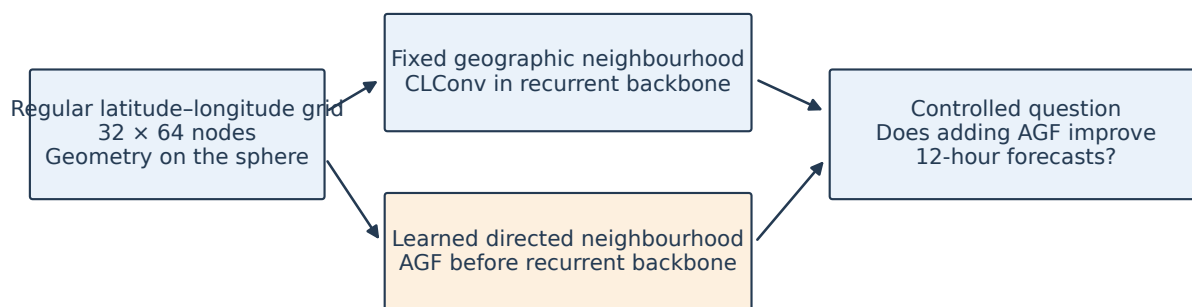


Figure S1: Scope of the controlled comparison. The fixed geographic neighbourhood is used by CLConv in both models. CLCRN-AGF adds a learned directed neighbourhood before the recurrent backbone. This conceptual diagram makes no claim that either neighbourhood identifies a particular physical process or that a measured form of geometric overfitting is reduced.

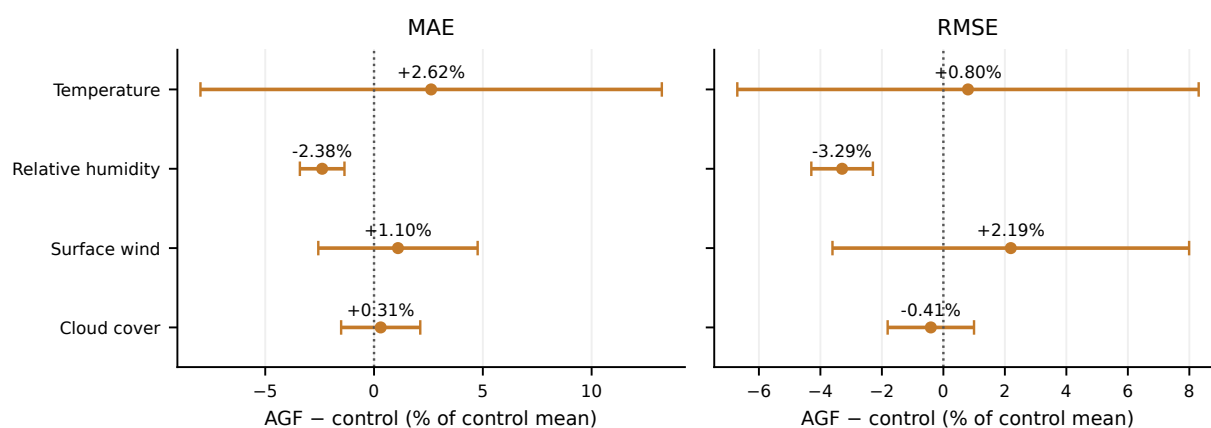


Figure S2: AGF-minus-control mean differences and 95% Welch intervals from Table S9, divided by the corresponding observed control mean and expressed as percentages. Negative values favour AGF. These are rescaled difference intervals, not confidence intervals from a separate ratio estimator. No single selected AGF run or historical schedule result enters this figure.

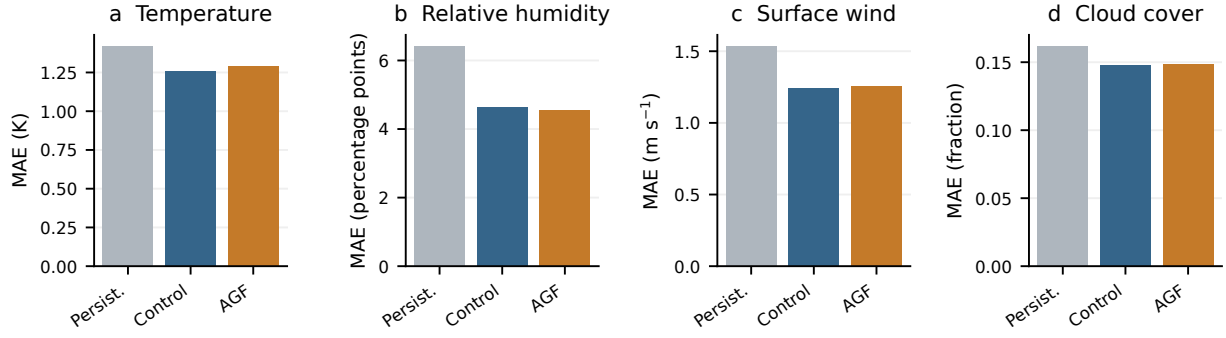


Figure S3: Current-split cumulative 12-hour MAE. Persistence is deterministic; control and AGF are five-seed means. All results use the same zero-inclusive, spatially unweighted evaluator. Panels have task-specific physical units. This is contextual, not a causal decomposition of schedule changes.

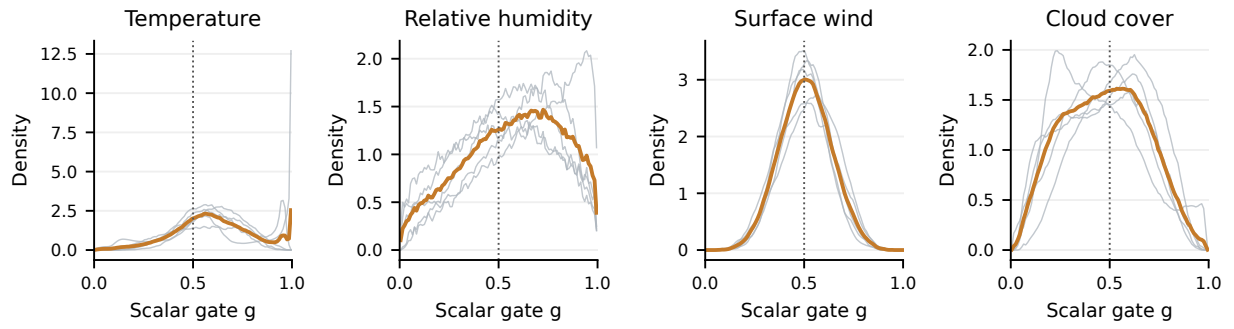


Figure S4: Full-test distributions of scalar sigmoid gates for all four tasks and the same five AGF checkpoints as the primary model tables above. Gates are collected over 657 test windows, 12 input hours, 2048 nodes and 8 output channels per seed. Thin grey curves are individual seeds and the coloured curve is the pooled 100-bin density. The gate mixes adaptive graph aggregation and a nodewise residual projection before CLConv. Gate magnitudes do not by themselves measure the relative contribution of the two branches.

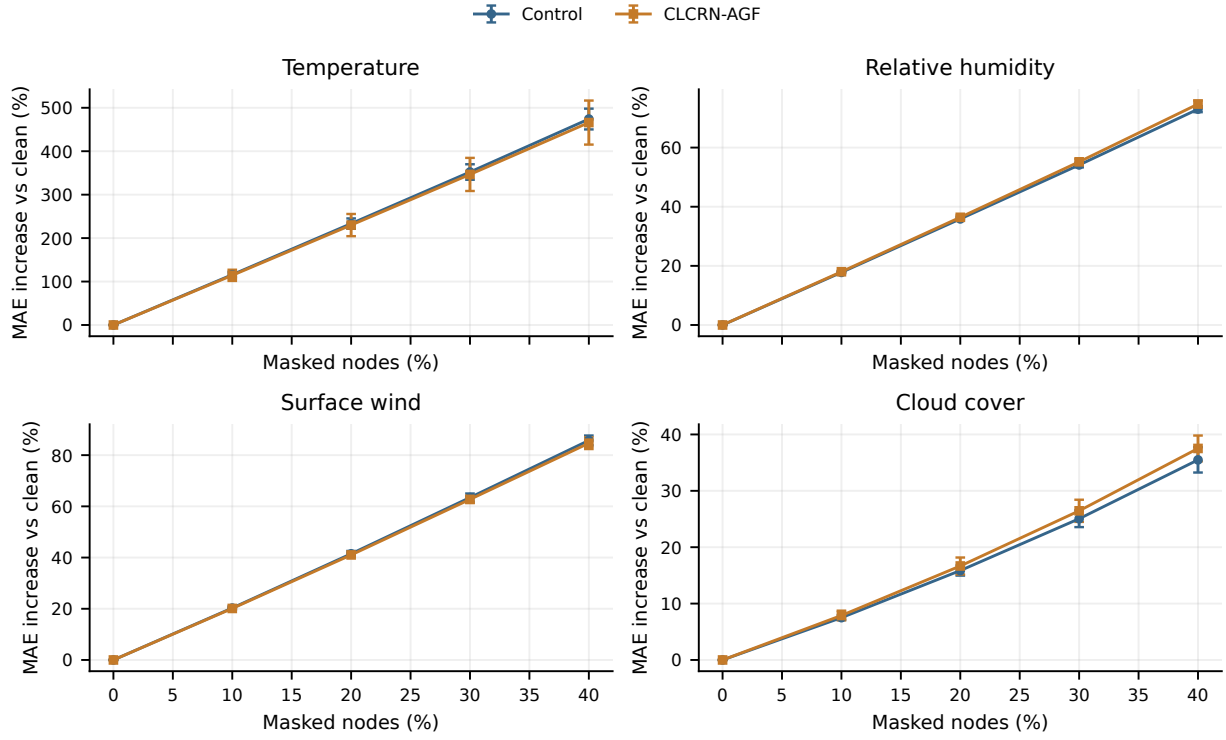


Figure S5: Control and AGF missing-input sensitivity corresponding to the AGF curves in main Figure 7 and to Table S14. Each checkpoint is evaluated on the same 64 deterministically spaced test windows. At each nonzero ratio, node masks from seeds 11, 22 and 33 are held fixed across all input hours and signal channels, and standardized inputs at masked nodes are set to zero. Relative MAE changes use each checkpoint's clean score on the same subset. Mask repeats are averaged within a training seed before calculating the mean and sample standard deviation across five training seeds. Error bars describe seed variation; they are not confidence intervals or a robustness significance test.

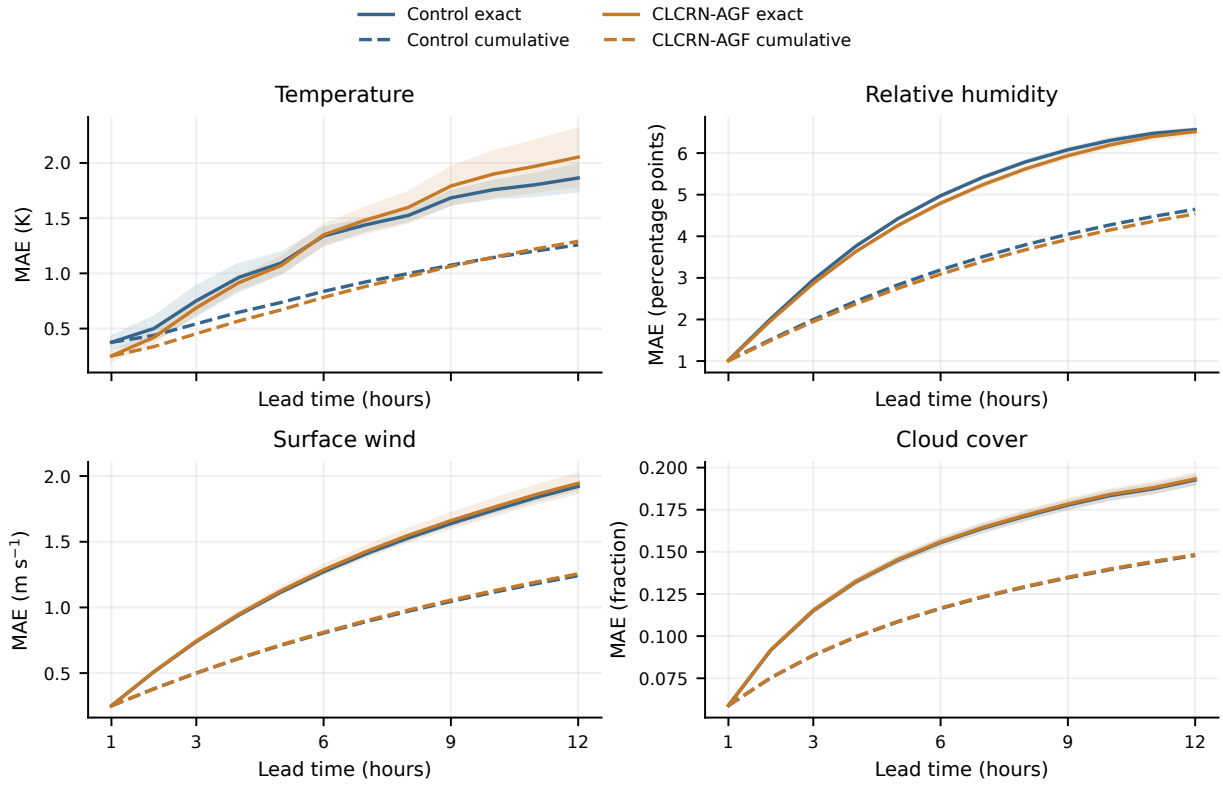


Figure S6: Exact-lead and cumulative MAE for control and AGF, averaged over the same five seeds per variant as the primary tables. Exact-lead scores use lead h only, while cumulative scores pool leads 1 through h . Shading denotes one sample standard deviation for exact-lead MAE. All models use all 657 test windows and the common zero-inclusive, spatially unweighted evaluator. Cumulative endpoints at 12 hours agree with Table S9. Main Figure 5 shows only the AGF models and adds exact-lead RMSE; this supplementary figure preserves the direct control–AGF MAE comparison.